MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 4C35A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figure 1

Mounting position: Any

Weight: 8-ounces (226.8 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ecc	egx	egy
Unit:	V	kv	kv	Vdc	Vdc	v	v
Maximum:	6.6	8.0 (see note 1)	8.0 (see note 2)	---	---	200	---
Minimum:	5.7	---	5% epy	2,500	---	---	Note 3
TEST CONDITIONS:	6.3	8.0	---	---	0	---	130

ABSOLUTE RATINGS:

Parameter:	ib	Ib	prr	Pb	$\frac{dik}{dt}$	tk	TA
Unit:	a	mAdc	---	---	a/µs	---	°C
Maximum:	90	100	---	2.0×10^9	1,000	---	+90
Minimum:	---	---	---	---	---	180	-50
TEST CONDITIONS:	---	---	2,800	---	---	---	---

GENERAL:

Qualification - Required

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	5.50	6.70	Aac
3267	Instantaneous starting	5, 6	epy = 7.0 kv (min), tk = 180 sec; Ef = 6.60 Vac	0.65	II	---	---	---	---
3246	Operation (1)	5, 9	epy = 10.0 kv (min), Ef = 5.70 Vac, t = 300 sec	0.65	II	egy	---	130	v
3247	DC anode voltage for conduction	-	Ef = 5.70 Vac	0.65	II	Ebb	---	1,500	Vdc
3251	Pulse emission (method A)	-	ik = 90a (min); prr = 60 ±10%; tp = 5.0 µs ±10%; tr = 0.5 µs (max), specified time interval = 2.5 µs	0.65	II	egk	---	150	v
	Quality conformance inspection, part 2								
1031	High-frequency vibration	-	No voltages applied	---	---	---	---	---	---
3246	Operation (2)	-	Operation (1), Ef = 6.60 Vac	---	---	egy	---	130	v
3256	Anode delay time	-	Operation (1), Ef = 6.3 Vac; t = 120 sec	---	---	tad	---	0.60	µs
3256	Anode delay time drift	7	Anode delay time	---	---	Δtad	---	0.10	µs
3261	Time jitter	5	epy = 3.0 kv (max)	---	---	tj	---	0.005	µs
	Quality conformance inspection, part 3								
---	Life test	5	Group B, t = 96 hours "on" and 1 hour "off" (tube mounted horizontally), t = 500 hours	---	---	---	---	---	---
---	Life-test end points.								
3246	Operation (1) and operation (2)	8		---	---	egy	---	140	v
3247	DC anode voltage for conduction	-	egy = 140v	---	---	Ebb	---	2,000	Vdc
3256	Anode delay time	-	egy = 140v	---	---	tad	---	0.70	µs
3256	Anode delay time drift	-	egy = 140v	---	---	Δtad	---	0.10	µs
3261	Time jitter	-	egy = 140v	---	---	tj	---	0.01	µs
1031	Variable-frequency vibration	10	No voltages applied	---	---	---	---	---	---
1041	Shock	4,10	Angle = 24°	---	---	---	---	---	---
---	Electrical characteristics measured before and after shock	5,10	epy = 10.0 kv (min); t = 300 sec, epy = 3.0 kv (max)	---	---	egy Ebb tj	--- --- ---	130 1,500 0.005	v Vdc µs
(F) 3246	Operation at elevated ambient temperature	5,10	TA = 90°C; Ef = 6.30 ±0.50 Vac; t = 5.0 hours	---	---	egy	---	130	v

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NOTES:

1. For instantaneous starting applications where anode voltage is applied instantaneously, the maximum permissible epy is 7.0 kv and shall be attained in not less than 0.04 second.

2. In pulsed operation, the peak inverse voltage, exclusive of spike of 0.05 µs maximum duration, shall not exceed 2.5 kv during the first 25 µs after the pulse.

3. The driver pulse, measured at tube socket with thyratron grid disconnected, shall be: egy = 175 v minimum, time of rise = 0.5 µs maximum, grid pulse duration = 2 µs minimum. Impedance of drive circuit = 1,500-ohms maximum.

4. Use clamp as specified on Drawing 243-JAN.

5. The anode circuit constants shall be so chosen that at epy = 8.0 kv under resonant charging conditions: dik/dt = 1,000 a/µs minimum; ib = 90 a; tp = 0.5 µs ±10 percent; prr = 3,000. The grid pulse characteristic shall be tp = 2.0 µs maximum; tr = 0.5 µs minimum; driver impedance = 1,500 ohms minimum.

6. The grid-drive voltage shall be applied within 10 seconds before application of anode voltage. The tube shall operate satisfactorily on pushbutton starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 7,000 v within 0.03 second (the filter in the rectifier shall be so designed that the epy reaches at least 3,500 v within 0.015 second). Any tube failing to start within three attempts shall be considered a failure.

7. During the interval between 2 minutes and 7 minutes of the anode delay time test, the change in anode delay time (Δtad) relative to the tad value observed on the anode delay time test shall not exceed the specified value.

8. Anode heating shall not be cause for rejection on operation (1) and operation (2) performed during life-test end point tests.

9. This test shall be the first test performed at the conclusion of the holding period.

(F)10. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

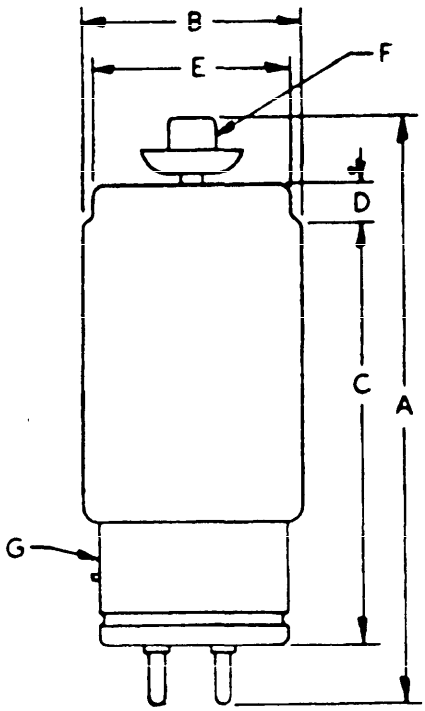
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2936)

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PIN CONNECTIONS	
Pin No.	Element
1	g
2	h, k
3	h
4	k
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.250 (158.75)	6.875 (174.63)
B	2.380 (60.45)	2.562 (65.07)
Quality conformance inspection, part 3 (see note)		
C	4.000 (101.60)	5.000 (127.00)
D	.400 (10.16)	---
E	2.000 (50.80)	2.300 (58.42)
F	Cap: C1-43 (EIA)	
G	Base: A4-18 (EIA)	

F

NOTE:
These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable failures
n2 = 4 c2 = 1; for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 basis.

FIGURE 1. Outline drawing of electron tube type 4C35A.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the s ification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for taining information on the use of this specification which will insure that suitable products can be procured wit minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold lines or reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 1

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